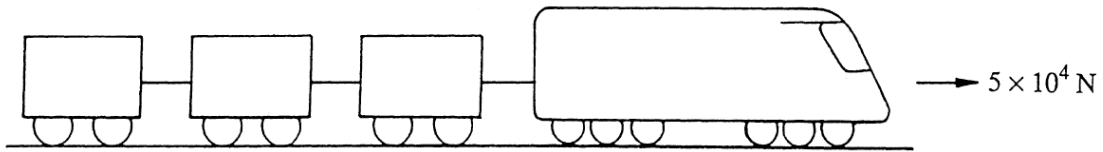


Warm up: trains

1



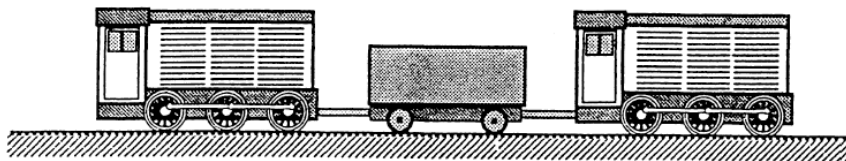
The diagram shows a goods train consisting of an engine of mass 40 tonnes and three trucks of 20 tonnes each. The engine is producing a driving force of 5×10^4 N, causing the train to accelerate. The ground is level and resistance forces may be neglected.

- (i) Find the acceleration of the train. [3]
- (ii) Draw a diagram to show the forces acting on the truck next to the engine. [2]
- (iii) Find the tensions in each of the three couplings. [3]

The brakes on the middle truck are faulty and suddenly engage, causing a resistance of 10^4 N.

- (iv) What effect does this have on the tension in the coupling to the last truck? [2]

2



A short train consists of two locomotives, each of mass 20 tonnes, with a truck of mass 10 tonnes coupled between them, as shown in the diagram. The resistances to forward motion are 0.5 kN on the truck and 1 kN on each of the locomotives. The train is travelling along a straight, horizontal section of track.

Initially there is a driving force of 15 kN from the front locomotive only.

- (i) Calculate the acceleration of the train. [4]
- (ii) Draw a diagram indicating the horizontal forces acting on each part of the train, including the forces in each of the couplings. Calculate the forces acting on the truck due to each coupling. [5]

On another occasion each of the locomotives produces a driving force of 7.5 kN in the same direction and the resistances remain as before.

- (iii) Find the acceleration of the train and the forces now acting on the truck due to each of the couplings. Compare your answer to this part with your answer to part (ii) and comment briefly. [5]